Circles

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In ΔABC studying inequalities in one triangle, $m\angle A = 43^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 85°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (43^\circ + 52^\circ) = 85^\circ$.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

In ΔABC studying the triangle inequality, $m\angle A = 44^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 91°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (44^\circ + 45^\circ) = 91^\circ$.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Given $f(x) = x^2 - 4x + 7$ with restricted domain $x \geq 2$, find an explicit formula for $f^{-1}(x)$ and state its domain.

Step 1: Complete the square to isolate x: $y = (x^2 - 4x + 4) + 3 = (x - 2)^2 + 3$. Step 2: Swap x and y (with $y \geq 2$): $x = (y - 2)^2 + 3 \Rightarrow (y - 2)^2 = x - 3$. Step 3: Take the principal square root (since $y \geq 2 \Rightarrow y - 2 \geq 0$): $y - 2 = +\sqrt{x - 3} \Rightarrow y = 2 + \sqrt{x - 3}$. Therefore, $\mathbf{f^{-1}(x) = 2 + \sqrt{x - 3}}$, defined for domain $\mathbf{x \geq 3}$ (which matches the range of f).

4 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

An investment fund gains 30% in value during Year 1, decreases by 20% in Year 2, and increases by 10% in Year 3. Determine the overall net percent change of the fund across the three-year period.

Step 1: Express each yearly change as a decimal multiplier:Year 1 (+30%): $1 + 0.30 = 1.30$Year 2 (-20%): $1 - 0.20 = 0.80$Year 3 (+10%): $1 + 0.10 = 1.10$ Step 2: Multiply the factors sequentially: $M_{\text{net}} = 1.30 \times 0.80 \times 1.10 = 1.04 \times 1.10 = 1.144$. Step 3: Convert to net percentage change: $\text{Net Percent Change} = (1.144 - 1) \times 100\% = \mathbf{+14.4\%}$. The investment gained an overall 14.4%.

5 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Use the quadratic formula to solve $x^2 + 2x - 8 = 0$.

Step 1: Identify coefficients: $a = 1$, $b = 2$, $c = -8$. Step 2: Apply the quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-2 \pm \sqrt{2^2 - 4(1)(-8)}}{2(1)} = \frac{-2 \pm \sqrt{4 + 32}}{2} = \frac{-2 \pm \sqrt{36}}{2}$ Step 3: Evaluate roots: $x = \frac{-2 \pm 6}{2} \Rightarrow x_1 = \frac{4}{2} = \mathbf{2}, \quad x_2 = \frac{-8}{2} = \mathbf{-4}$ The solutions are $\mathbf{x = 2}$ and $\mathbf{x = -4}$.

© ClipSAT — clipsat.org | All Rights Reserved

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A fair coin is tossed twice. Find $P(\text{two heads})$.

Because each coin flip is an independent event, apply the multiplication rule for independent probabilities: $P(\text{Head on 1st}) = \frac{1}{2}$ $P(\text{Head on 2nd}) = \frac{1}{2}$ $P(H_1 \cap H_2) = P(H_1) \times P(H_2) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4} = \mathbf{0.25}$ Sample Space Check: The sample space is $\{HH, HT, TH, TT\}$. Exactly 1 outcome out of 4 is HH, confirming $\frac{1}{4}$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A solid toy is formed by attaching a cone of radius 4 cm and height 9 cm to the flat face of a solid hemisphere of radius 4 cm. Calculate the total volume of the toy. Give your answer correct to 3 significant figures.

Step 1: Volume of cone: $V_{\text{cone}} = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(4^2)(9) = 48\pi \text{ cm}^3$ Step 2: Volume of hemisphere: $V_{\text{hemi}} = \frac{1}{2} \times \frac{4}{3}\pi r^3 = \frac{2}{3}\pi(4^3) = \frac{128}{3}\pi \text{ cm}^3$ Step 3: Total volume:

$V_{\text{total}} = 48\pi + \frac{128}{3}\pi = \frac{272}{3}\pi \approx 284.838\ldots \approx 285 \text{ cm}^3$ (3 s. f.)

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A school theater sold 250 tickets for a play, raising a total of $1,750$. $Adulttickets cost 9$ each and student tickets cost 5 each. How many adult tickets were sold?

Set up system of equations: Let a be the number of adult tickets and s be student tickets: $\{a + s = 250; 9a + 5s = 1750$ Elimination method: Multiply the first equation by 5: $5a + 5s = 1250$. Subtract from the second equation: $(9a + 5s) - (5a + 5s) = 1750 - 1250 \Rightarrow 4a = 500 \Rightarrow \mathbf{a = 125}$. Exactly 125 adult tickets were sold (and $s = 250 - 125 = 125$ student tickets).

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A recipe is scaled from 2 to 5 servings. How many cups replace 4 cups?

Set up a direct proportion where x represents the number of cups for 5 servings: $\frac{2 \text{ servings}}{4 \text{ cups}} = \frac{5 \text{ servings}}{x \text{ cups}}$ Cross-multiply to solve for x: $2x = 5 \times 4 = 20 \Rightarrow x = \frac{20}{2} = \mathbf{10}$ cups Alternatively, calculate the scaling factor: $k = \frac{5}{2} = 2.5$. Then $4 \times 2.5 = \mathbf{10}$ cups.

© ClipSAT — clipsat.org | All Rights Reserved

10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Determine the equation of the circle in the xy-plane for which the segment connecting $A(2, -3)$ and $B(8, 5)$ serves as a diameter.

Step 1: The center (h, k) is the midpoint of diameter AB: $(h, k) = \left(\frac{2+8}{2}, \frac{-3+5}{2}\right) = \left(\frac{10}{2}, \frac{2}{2}\right) = \mathbf{(5, 1)}$. Step 2: Compute radius r as the distance from center $(5, 1)$ to $B(8, 5)$:

$r^2 = (8-5)^2 + (5-1)^2 = 3^2 + 4^2 = 9 + 16 = 25 \Rightarrow r = 5.$ Step 3: Write standard equation:

$\mathbf{(x-5)^2 + (y-1)^2 = 25}$.

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Level 3 · AdvancedWhat is the set of all real solutions to the radical equation $\sqrt{2x-5} = x-4$?

Square both sides: $2x - 5 = (x-4)^2 = x^2 - 8x + 16 \Rightarrow x^2 - 10x + 21 = 0 \Rightarrow (x-3)(x-7) = 0.$ Check prospective roots in original equation:For $x = 3$: $\sqrt{2(3)-5} = \sqrt{1} = 1$, but right side is $3 - 4 = -1$. Since $1eq -1$, $x = 3$ is extraneous.For $x = 7$: $\sqrt{2(7)-5} = \sqrt{9} = 3$, and right side is $7 - 4 = 3$. Valid!The only real solution is $x = 7$.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Convert the quadratic function $y = 2x^2 - 12x + 11$ into vertex form $y = a(x-h)^2 + k$. State the coordinates of the vertex and the line of symmetry.

Step 1: Factor the leading coefficient $a = 2$ from the x-terms: $y = 2(x^2 - 6x) + 11$ Step 2: Complete the square inside parentheses. Half of -6 is -3, and $(-3)^2 = 9$:

$y = 2(x^2 - 6x + 9 - 9) + 11 = 2(x-3)^2 - 2(9) + 11 = 2(x-3)^2 - 18 + 11$ Step 3: Simplify to vertex form: $y = \mathbf{2(x-3)^2 - 7}$ The vertex is $\mathbf{(3, -7)}$ and the axis of symmetry is $\mathbf{x = 3}$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A bag contains 5 red marbles, 4 blue marbles, and 3 green marbles. Two marbles are drawn at random one after another without replacement. What is the probability that both marbles are red?

Total number of marbles initially: $5 + 4 + 3 = 12$. Probability that the first marble is red: $P(R_1) = \frac{5}{12}$. Since it is drawn without replacement, 11 marbles remain, 4 of which are red. Probability that the second marble is red: $P(R_2|R_1) = \frac{4}{11}$. Using the multiplication rule: $P(R_1 \cap R_2) = \frac{5}{12} \times \frac{4}{11} = \frac{20}{132} = \mathbf{\frac{5}{33}}$

© ClipSAT — clipsat.org | All Rights Reserved

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The scores on a statewide standardized test are normally distributed with a mean of $\mu = 500$ and a standard deviation of $\sigma = 100$. Approximately what percentage of test takers score between 400 and 700?

Express the boundaries in terms of standard deviations from the mean: $400 = \mu - 1\sigma$ $700 = \mu + 2\sigma$ Using the Empirical Rule: Between $\mu - 1\sigma$ and μ: 34.1%. Between μ and $\mu + 1\sigma$: 34.1%. Between $\mu + 1\sigma$ and $\mu + 2\sigma$: 13.6%. Total percentage: $34.1\% + 34.1\% + 13.6\% =$ **81.8%** (or approximately **$81.5\% - 82\%$**).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 11.01 Probability

A fair die is rolled twice. Find $P(\text{both show } 6)$.

Step 1: Determine independent probabilities: For a fair six-sided die, the probability of rolling a 6 on any single roll is $P(6) = \frac{1}{6}$. Step 2: Apply the multiplication rule for independent events: $P(\text{both } 6) = P(6) \times P(6) = \frac{1}{6} \times \frac{1}{6} = \mathbf{\frac{1}{36}} \approx 0.0278$ The probability is $\mathbf{\frac{1}{36}}$ (or 1 out of 36 outcomes).

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Lines & Angles

A square has area 36. Find its perimeter.

Step 1: Find the side length s from area $A = s^2$: $s = \sqrt{36} = 6$ Step 2: Calculate the perimeter of the square $P = 4s$: $P = 4 \times 6 =$ **24** Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Triangles & Congruence

Suppose y varies directly as x^2 and inversely as z. If $y = 18$ when $x = 3$ and $z = 4$, find the value of y when $x = 5$ and $z = 10$.

Step 1: Formulate the joint variation equation: $y = k\frac{x^2}{z}$ Step 2: Solve for constant k using the given initial condition: $18 = k\frac{3^2}{4} = k\frac{9}{4} \Rightarrow k = 18 \times \frac{4}{9} = 2 \times 4 = 8$. Step 3: Calculate y for $x = 5$ and $z = 10$: $y = 8\frac{5^2}{10} = 8\frac{25}{10} = 8 \times 2.5 =$ **20**. The value of y is **20**.

© ClipSAT — clipsat.org | All Rights Reserved

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

A car travels at 60 mph for 2.5 hours. Find the distance.

Apply the uniform distance formula $d = r \cdot t$: $d = 60 \text{ miles/hour} \times 2.5 \text{ hours} = 60 \times \frac{5}{2} = \mathbf{150}$ miles Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Right Triangles & Trigonometry

A jacket costs USD 50 after a 20% discount. Find the original price.

Let P be the original price. A 20% discount leaves $100\% - 20\% = 80\%$ of the original price: $0.80 \times P = 50$ $P = \frac{50}{0.80} = \frac{50}{4/5} = 50 \times \frac{5}{4} = \frac{250}{4} = \mathbf{USD62.50}$ Trap Warning: Do NOT take 20% of $USD50$ and add it ($USD50 + USD10 = USD60$); percentages must always be calculated relative to the original base!

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Polygons & Quadrilaterals

Solve the equation $5e^{0.04t} = 45$ for t in terms of natural logarithms, and find its value rounded to two decimal places.

Step 1: Isolate the exponential expression: $e^{0.04t} = \frac{45}{5} = 9$. Step 2: Take the natural logarithm of both sides: $\ln(e^{0.04t}) = \ln(9) \Rightarrow 0.04t = \ln(9)$. Step 3: Solve for t: $t = \frac{\ln(9)}{0.04} = 25\ln(9) = 25\ln(3^2) = 50\ln(3) \approx 50(1.0986) = \mathbf{54.93}$. The exact solution is $\mathbf{50}\ln\mathbf{(3)}$ (approximately 54.93).

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying angle bisectors and perpendicular bisectors, $m\angle A = 41^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 94° ✓

b. 104°

c. 86°

d. 139°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (41^\circ + 45^\circ) = 94^\circ$.

© ClipSAT — clipsat.org | All Rights Reserved

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying altitudes, medians, and bisectors of triangles, $m\angle A = 42^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 93° ✓

b. 103°

c. 85°

d. 138°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (42^\circ + 45^\circ) = 93^\circ$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 3x + 9$ in dividing a segment into a given ratio, evaluate $f(4)$.

a. 21 ✓

b. 24

c. 12

d. 25

Substitute $x = 4$: $f(4) = 3(4) + 9 = 21$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Two chords intersect inside a circle. One chord is divided into segments of length 5 and 8; the other into segments of length 10 and x. Find x.

a. 10

b. **4** ✓

c. 6

d. 13

By the Intersecting Chords Theorem, $5 \times 8 = 10 \times x$, so $x = \frac{5 \times 8}{10} = 4$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A central angle in a circle measures 215°. What is the measure of its intercepted arc?

a. 35°

b. 107°

c. 145°

d. **215°** ✓

A central angle's measure equals the measure of its intercepted arc: 215°.

© ClipSAT — clipsat.org | All Rights Reserved

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

An inscribed angle intercepts an arc of 144°. What is the measure of the inscribed angle?

a. 144°

b. 108°

c. 216°

d. **72°** ✓

An inscribed angle is half the intercepted arc: $144/2 = 72^\circ$.

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A circle has radius 11. Find the area of a sector with central angle 120°, in terms of π.

a. $\frac{121}{3}\pi$ ✓

b. $\frac{122}{3}\pi$

c. 121π

d. $\frac{123}{3}\pi$

Sector area $= \frac{120}{360} \times \pi(11)^2 = \frac{121}{3}\pi$.

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A circle has radius 11. Find the arc length of a sector with central angle 120°, in terms of π.

a. 22π

b. 11π

c. $\frac{23}{3}\pi$

d. $\frac{22}{3}\pi$ ✓

Arc length $= \frac{120}{360} \times 2\pi(11) = \frac{22}{3}\pi$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

© ClipSAT — clipsat.org | All Rights Reserved

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In the xy-plane, the circle given by the equation $x^2 + y^2 - 4x + 6y - 12 = 0$ is tangent to line ℓ at the point $(5, 1)$. Which of the following is an equation of line ℓ?

a. $3x + 4y = 19$ ✓

b. $4x + 3y = 23$

c. $3x - 4y = 11$

d. $4x - 3y = 17$

To find the equation of the tangent line ℓ, we first find the center of the circle by completing the square: 1. Group the x and y terms: $(x^2 - 4x) + (y^2 + 6y) = 12$ 2. Complete the square for both variables: $(x^2 - 4x + 4) + (y^2 + 6y + 9) = 12 + 4 + 9$ $(x - 2)^2 + (y + 3)^2 = 25$ Thus, the center of the circle is $C(2, -3)$ and the radius is $r = 5$. 3. Determine the slope of the radius connecting the center $C(2, -3)$ to the point of tangency $P(5, 1)$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In the circle shown, chords AB and CD intersect at point P. If $PA = 6$, $PB = 8$, and $PC = 4$, what is the value of x, the length of segment PD?

a. 10

b. **12** ✓

c. 16

d. 18

By the Intersecting Chords Theorem (Power of a Point inside a circle), when two chords AB and CD intersect at point P, the products of the lengths of their segments are equal: $PA \cdot PB = PC \cdot PD$ Substitute the given values into the equation: $6 \cdot 8 = 4 \cdot x$ $48 = 4x$ $x = 12$ Therefore, the length of segment PD is 12. Misconception Analysis: - Choice A (10): Additive error, incorrectly assuming segment lengths add across chords ($6 + 8 = 4 + x$, yielding $x = 10$). - Choice C (16): Finds.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Point $(5, -1)$ is dilated about the origin with scale factor 2. What are the coordinates of the image?

a. **(10, -2)** ✓

b. (10, -1)

c. (7, 1)

d. (-10, 2)

A dilation about the origin with scale factor k maps $(x, y) \to (kx, ky)$: $(2 \times 5, 2 \times -1) = (10, -2)$.

© ClipSAT — clipsat.org | All Rights Reserved

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A rectangle has length 7 and width 6. What is its perimeter?

a. 30

b. **26** ✓

c. 13

d. 42

Applying Area & Perimeter principles, Perimeter $= 2(l + w) = 2(7 + 6) = 26$. Therefore, the verified correct option is 26.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A regular 4-gon has side length 2 and apothem 11. What is its area?

a. 88

b. **44** ✓

c. 8

d. 50

Area $= \frac{1}{2} \times$ perimeter $\times$ apothem $= \frac{1}{2} \times (8) \times (11) = 44$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A test consists of 25 multiple-choice questions. A student scored 80%. If the student correctly answered 4 more questions, what would the new percentage score be?

a. 84%

b. 88%

c. 92%

d. **96%** ✓

Initial correct: $0.80 \times 25 = 20$. New correct: $20 + 4 = 24$. New score: $24/25 = 96\%$.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A triangle has sides 75, 130, and 185. Using Heron's formula, find its area.

a. 2240

b. **2250** ✓

c. 390

d. 2265

Semi-perimeter $s = \frac{75 + 130 + 185}{2} = 195$. Area $= \sqrt{s(s-a)(s-b)(s-c)} = 2250$.

© ClipSAT — clipsat.org | All Rights Reserved

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Point $(4, -10)$ is reflected over the line $y = x$. What are the coordinates of the image?

a. (-4, -10)

b. (10, -4)

c. **(-10, 4)** ✓

d. (4, -10)

Reflecting over $y = x$ swaps the coordinates: $(4, -10) \to (-10, 4)$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A rectangular prism has length 3, width 3, and height 10. What is its surface area?

a. 16

b. 146

c. 90

d. **138** ✓

Surface area $= 2(lw + wh + lh) = 2(3 \times 3 + 3 \times 10 + 3 \times 10) = 138$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

Total surface area of a cylinder with radius 3 and height 5 (in terms of π):

a. 57π

b. 30π

c. 15π

d. 48π ✓

$SA = 2\pi r^2 + 2\pi rh = 2\pi r(r + h) = 2\pi \times 3 \times (3 + 5) = 48\pi$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Right Triangles & Trigonometry

A cylinder has radius 5 and height 18. Find its total surface area in terms of π.

a. 180π

b. 450π

c. 239π

d. 230π ✓

Surface area $= 2\pi r^2 + 2\pi rh = 2\pi(5)^2 + 2\pi(5)(18) = 230\pi$.

© ClipSAT — clipsat.org | All Rights Reserved

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Polygons & Quadrilaterals

A cone has radius 7 and height 24. Find its volume in terms of π.

a. 168π

b. 397π

c. 392π ✓

d. 1176π

Volume $= \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(7)^2(24) = 392\pi$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Area, Surface Area & Volume

Volume of a cone with radius 3 and height 4 (in terms of π):

a. 36π

b. 12π ✓

c. 19π

d. 4π

$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi \times 3^2 \times 4 = 12\pi$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 9.01 Transformations & Symmetry

A sphere has radius 12. Find its volume in terms of π.

a. 1728π

b. 2311π

c. 576π

d. 2304π ✓

Volume $= \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(12)^3 = 2304\pi$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Foundations & Reasoning

A trapezoid has parallel sides 4 and 11 and height 20. What is its area?

a. 35

b. 162

c. 300

d. **150** ✓

Area $= \frac{1}{2}(b_1 + b_2)h = \frac{1}{2}(4 + 11)(20) = 150$.

© ClipSAT — clipsat.org | All Rights Reserved

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 10.01 Coordinate Geometry

A rectangular pyramid has a base of length 2 and width 12, and a height of 5. What is its volume?

a. 120

b. 48

c. **40** ✓

d. 19

Volume $= \frac{1}{3}lwh = \frac{1}{3}(2)(12)(5) = 40$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 11.01 Probability

A triangle has base 6 and height 9. What is its area?

a. 30

b. 54

c. **27** ✓

d. 15

Area $= \frac{1}{2}bh = \frac{1}{2}(6)(9) = 27$.

© ClipSAT — clipsat.org | All Rights Reserved